Reflection on the use of (model-based) knowledge in the negotiations

EPA141A
Group 1

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June 20, 2025

Introduction

Flood risk management is a multi-layered and complex problem in The Netherlands. It is characterized by interacting elements surrounded by uncertainties, long time frames, conflicting views among stakeholders and therefore subject to deep uncertainty and problem framing. During the *Room for the River* project negotiations, even though several actors had claims that we thought of as rational and logical, the wickedness of this case made framing contestable. Wicked problems are societal issues that lack definite solutions because decisions involve many stakeholders with differing values and interests (Rittel & Webber, 1973). For example, the Transport Company's (TC) objective - avoiding policies that could decrease the original minimum water level to prevent the fluvial transport economy in The Netherlands from decreasing- was not seen as a priority for most actors. Hence, the TC faced a hard time proving and making its role relevant for the authorities and other actors. In these situations, scientific information can easily become politicized.

Science can provide opposing parties with "their own bodies of relevant, legitimated facts about nature", selected partly because they align with their specific interests and beliefs (Sarewitz, 2004).

In short, beyond rationality, there was a political game to be played. This surprised us as analysts and gave us relevant takeaways that will be explained.

In this section we will address the challenges encountered during the decision-making process for the *Room for the River* project, how knowledge was used and the (potential) role of model-based approaches.

Identification of roles of knowledge

The *Room for the River* project and its decision-making process have been reproduced as an exercise to put in practice the roles of knowledge from the academic literature. In this part we introduce four types along with a specific example from the debate.

First, the so-called *Speaking Truth to Power* is a traditional, "old-fashioned idea" posits that scientific facts provide an "appropriate foundation for knowing how to act in the world" and should "settle disputes and guide political action (Sarewitz, 2004). This approach, often termed the "predict-and-act" approach, aims to incorporate all known facts into a "best estimate package" for predictive system behavior, relying on a single best estimate of the future (Moallemi et al., 2020). The actor that did this most during the debate was the Delta Commission. They argued for policies that were justified by favourable outcomes in the model. Similarly, Rijkswaterstaat attempted to reach consensus by gathering all needs in a matrix in the first high level debate.

Second, *Muddling Through* is a role of knowledge identified by Lindblom (1959) which relies heavily on the record of past experience with small policy steps to predict the consequences of similar steps extended into the future. This means that knowledge is built incrementally from observed outcomes of previous actions. For instance, during the negotiations Rijkswaterstaat adapted the proposed policy multiple times after receiving feedback by email. They did this by doing small incremental fixes to their original proposal. They were learning about other stakeholders' as the negotiation process was going along.

Additionally, the policy levers, such as RfR and dike heightening (DH) are also an example of Muddling Through. These are policies that can be implemented upon the existing infrastructure. For instance, DH is only considered in areas where a dike already exists. As a result, more transformative options, such as a full dike relocation, are excluded from the solution space, even though they might be more optimal.

Thirdly, *Politics of Knowledge* consist of leveraging and ignoring knowledge strategically to support an already pre-defined (set of) ideas and objectives in the political game. This took place as well during the debate. Rijkswaterstaat first didn't want to incorporate any other outcome measures than the one that were already in the model. However, Minimum Water Level, which was a new outcome, was the main objective of the TC. Rijkswaterstaat tried to "ignore" this knowledge in the beginning.

This behaviour aligns with the understanding that science provides parties with their own validated facts, which are often chosen because they support specific interests (Sarewitz, 2004).

Lastly, Rittel and Webber (1973) defined wicked problems as those which have no definitive formulation or solution, and their resolution depends heavily on the values and interests of the many stakeholders involved. They are systems of problems rather than isolated ones (Ackoff, 1979). In other words, a great part of its complexity relies on the fact that the mere problem formulation can be contested. Gelderland questioned everything because the model did not take into account land as a metric, which was important for them, and that the current direction of the negotiations was not ethical nor effective. The problem was not necessarily a lack of scientific knowledge, but an "excess of objectivity", where knowledge can be interpreted in different ways, leading to competing views of the problem itself (Sarewitz, 2004).

During the student-led exercise, we observed the four roles put into action by several actors and its consequences. Our position as analysts provided us a privileged perspective to analyze the main actors' negotiating techniques. After briefly introducing these concepts, we will delve deeper into the consequences of using them in the next section.

Systematic analysis of debate

In general, we perceived that the model played a key role in the debate, being the main source of knowledge for all actors. This is not the normal case. Not all actors are knowledgeable on modelling or even opt for a model-based decision making approach in the first place. Nevertheless, the educational context in which this discussion takes place, TU Delft (engineers), makes this a bit different.

While we will elaborate on this rational, model-based approach, we acknowledge the boundaries of the educational context and it only partially reflects the real-world.

First round of debates

As stated before, in the first high level debate, Rijkswaterstaat took a rational approach to policy-making (Kwakkel et al., 2016) which also aligns with the traditional role of knowledge, speaking truth to power. Rijkswaterstaat led the debate by creating a matrix outcomes/actors to determine which outcomes were of major priority for all actors. By this, they intended to identify a few trade-offs, reduce the complexity of the problem, to use in the model and find an optimal solution. Below, table 1 illustrates the matrix.

	Delta Commission	Environmental	Transport Company
Damage	4		5- “we cannot transport anymore and local economies are damaged”
Dike investment	2- They want to minimize dike investment in favour of Room for the River	5	1
# Deaths	5	4	3
R&D Costs	1	1	1
Evacuation costs	4		1

Table 1. Matrix and notes from the high-level debate

This method is a traditional “predict-and-act” approach. The matrix is used as a tool to aggregate the diverse perspectives into a shared, rational understanding of the problem (Sarewitz, 2004). Starting from this matrix, Rijkswaterstaat used the model to find an optimal policy. However, this approach is often inadequate for problems with deep uncertainty and conflicting values. It can limit the exploration of possibilities by trying to oversimplify the problem from the beginning (Walker et al., 2013). And it masks the complexity of the problem, and the underlying disagreements among stakeholders. Also, as Ackoff (1979) said: in situations of deep uncertainty the “optimal solution of a model is not an optimal solution of a problem unless the model is a perfect representation of the problem, which it never is”.

During the matrix analysis, another thing that stood out was Rijkswaterstaat’s response to the Transport Company’s (TC) concern about minimum water levels. TC emphasized this as their main priority, but Rijkswaterstaat dismissed it, arguing it was not in all actors’ models. This shows a missed opportunity to turn the Room for the River project into a multi-issue game (De Bruijn et al., 2015). Including TC could have broadened support from the beginning, especially since the status quo already benefits them. Instead, Rijkswaterstaat engaged in what Van Ernst et al. (2014) call the "strategic ignoring of knowledge”. This means disregarding input seen as inconvenient. Moreover, their reliance on the existing model reflects “excess of objectivity”. This term is used to describe situations where model outputs are treated as reality (Sarewitz, 2004). This narrowed the scope for discussion and produced slight tension in the encounter.

Moreover, Delta Commission called for transparency and demanded all models to be shared. Their goal was probably to promote better collaboration and a shared understanding of the models. However, this approach overlooks that a model is never a perfect representation of the problem, as previously discussed. This showed very clearly in the case of Gelderland. Their preferences, like the protection of farmers, could not be captured by the model. As a result, this undermined Gelderland legitimacy.

From the Overijssel debate, we saw from the start a disposition to collaborate with other provinces and higher authorities to get the most plausible outcome. For instance, the city of Deventer argued against heightening dikes despite lacking the land to do Room for the River and denying relocation as an option. This suggests a “Not in my Backyard approach” in which Overijssel actors seem collaborative for the RfR policies they expect Rijkswaterstaat to propose but in reality will sabotage them if it has to be in their location.

Lastly, the Gelderland debate was not possible to attend. Unlike previous debate moderators, this actor decided to have a closed doors debate with the dike rings. This limited the possibilities for external actors to understand the province's needs.

Second and Final round of debates

In the first debate, Gelderland didn't let any other actor listen to their discussion with their dike rings. They might have wanted this to be a strategic move to hold more information (although we aren't sure what their actual goals were). However, this might have worked against them and resulted in Gelderland not being able to satisfy their goals. As said before, due to Rijkswaterstaat's rational approach, they already worked on possible policies based on their matrix before the final debate. Therefore they couldn't know what Gelderland actually wanted.

Also the TC had the strategy to keep a lot of information to themselves in the beginning and mostly listen. As a result, they did not effectively advocate for or leverage their interests early on. So advocating for their interests during the second debate had less impact. Consequently, their goal of the minimum water level was not treated as a priority and addressed only through some additional, yet uncertain, measures.

Sources mention that when stakeholders withhold information or their interests are not included from the start, it blocks the important discussion needed to bring together different perspectives and make sense of complex and uncertain problems (Sarewitz, 2004). Also, when interests aren't included in the model because of missing information, it shows the need for what Rayner & Sarewitz (2021) call "cognitive pluralism" and trust building. Different actors view the problem differently and involving this diverse group of people can lead to shared understandings. These shared understandings might not be more correct, but can be more accepted.

In the final debate, Rijkswaterstaat's initial rational approach began to change. Whereas earlier they had tried to find an optimal solution through the model, they now started to move away from this strict "optimal policy". Instead, they used their policy as a starting point and listened to what actors wanted to change. Even when the proposed changes were not optimal according to the model, but also not "bad", they incorporated that change. This shows a shift from using a model as a predictive tool, to using a model as a "thinking aid" to explore, which aligns with the idea of exploratory modeling (Moallemi et al., 2020). This shift was also seen during the breaks of the debate. Rijkswaterstaat actively talked with all the actors to better understand their positions, aiming to make the debate into a multi-issue negotiation, rather than an optimization. This change of perspective made sense after they told us that their mandate was a 6/6 pass policy approval.

This shift of perspective, however, did not go far enough for the province of Gelderland. For them, Rijkswaterstaat still remained too focused on the model. As said previously, Gelderland's preferences could not be captured in the model and as a result this undermined their legitimacy and the final policy was not what they wanted. This outcome shows the difficulty of truly moving beyond that "excess of objectivity".

Our modeling choices based on the analysis

Our modelling decisions were shaped directly by the negotiation context and the position of the Transport Company, whose core was not included in the shared model during the early stages. This omission weakened their position in the debate, as their concern could not be assessed, compared, or negotiated alongside the interests of other actors. Without visibility in the model, the issue of water level risked remaining unquantified and, therefore, politically marginal.

We addressed this gap by incorporating minimum water level into the model outputs. This was not simply a technical modification, but a deliberate response to a common issue in model-based policy processes. We aim to avoid MWL from being filtered out because it does not align with dominant assumptions or existing metrics. When only a narrow set of outcomes is considered, concerns like those of the Transport Company are easily ignored. In such cases, actors may strategically ignore or exclude inconvenient knowledge to protect their preferred policy pathways (Van Ernst et al., 2014). By including a minimum water level, we challenged this selective framing and restored the relevance of our client's interest.

Once the minimum water level was added, we restructured the model to better reflect the multi-actor, multi-value nature of the negotiation. We included additional outcome measures valued by other stakeholders, such as *Expected Number of Deaths* and *Expected Annual Damage* for the Delta Commission and Rijkswaterstaat. Hence, we sharpened our modelling process with the exploratory role of knowledge. Rather than aiming to predict a single best policy, we used the model as a tool to explore a wide range of plausible futures, assumptions, and outcomes (Kwakkel et al., 2016b; Bankes, 1993). Exploratory modelling serves as a thinking aid that helps uncover trade-offs, tensions, and potential alignments between stakeholders under deep uncertainty.

As we analyzed the expanded model, our strategic perspective changed. Since the required minimum water level was satisfied in all scenarios, also the minimum needs of the Transport Company were satisfied. This allowed us to shift the aim of the analysis to consider the outcomes valued by other stakeholders, ensuring a policy that can gain broad support. Initially, we anticipated rejecting most Room for the River proposals. However, our exploration revealed that many policy combinations did not significantly affect minimum water levels. This insight allowed us to shift from opposition to selective engagement. We began identifying those few scenarios that would substantially harm our client's interests and focused on blocking only those. In robust decision-making the goal is not to find a single optimal solution but to identify options that perform well across a broad range of conditions (Walker et al., 2013). By identifying our "robust region" of acceptable outcomes, we were able to engage flexibly while protecting our red lines.

To support this, we performed a worst-case direct search to identify policy combinations that would cause unacceptable harm to the Transport Company. These scenarios became the boundaries of our negotiating space. The insights gained here were both strategic and analytical. Since a "No Deal" outcome remained acceptable for our client, we had leverage to reject certain proposals while remaining open to compromise. Scenario discovery of this kind is central to exploratory modelling and helps decision-makers understand under what conditions policies are likely to fail or succeed (Kwakkel et al., 2016a).

We settled on recommending only two distinct policies to our client. These policies showed the preferred levers of the transport company while offering two different options in terms of number of RfR projects, dike heightening extent, and implementation time steps. By recommending only two policies, we avoided overwhelming our client with information while still supplying all the relevant background information to provide insight into the recommended policies and support the transport company's position in the decision-making arena.

Instead of individual policy levers, the focus of the analysis was on complete policy sets. This approach increased the performance of the analysis by considering the complete system, including interaction effects between the policy levers. At the same time it also made the results more complex, reducing the suitability in a multi-actor arena where the focus is often on single policy levers instead of sets of levers. SOBOL sensitivity analysis provided insight into the effects of individual uncertainties but no similar approach was used for policy levers. In a river system with significant interaction effects, our approach to aggregate individual levers into policies is justified. Furthermore, all Pareto optimal policies were tested and the results included in the data files. In case needed, the results of any Pareto optimal policies can be looked at and compared against the recommended policies. The results of non-Pareto optimal policies are irrelevant for decision-making and thus excluded from detailed analysis since, by definition, these should not be preferred by any stakeholder when a more preferred Pareto optimal policy is available.

Naturally this is based on the assumption that all stakeholders are rational actors aiming to maximize benefit.

In summary, our modelling choices reflected a deliberate use of exploratory modelling to manage deep uncertainty, align with stakeholder diversity, and support robust negotiation. By expanding the outcome space, focusing on performance across scenarios, and adjusting our stance based on new insights, we used the model as more than a forecasting tool. It became a political and strategic platform that enabled the Transport Company to participate meaningfully in a contested and evolving policy space.

Recommendations to actors

In the previous sections, several challenges and shortcomings in the decision-making process were identified. This section focuses on addressing the main challenges and gives recommendations on how to tackle them specific to the policy advice provided in the prior report.

As a reminder of the previous report´s results, the proposed policies the TC should advocate for are provided below:

Policy 2:

- Two room for the river projects in Gelderland: Olburgen and Tichelbeeksewaard
- Two dike increase locations in Gelderland: Cortenoever and Zutphen
- Two dike increase locations in Overijssel: Gorssel and Deventer

Policy 9:

- Three room for the river projects in Gelderland: Olburgen, Tichelbeeksewaard, and Obstakelverwijdering.
- Two Dike increase locations in Gelderland: Cortenoever and Zutphen
- Two Dike increase locations in Overijssel: Gorssel and Deventer

While these proposals are rational and provide a desirable trade-off between MWL, EAD and expected number of deaths, there are more factors to take into account in the political arena. These policies do not equally distribute its burdens. All rooms for the river projects are in the province of Gelderland, which is heavily reliant on agriculture and will struggle if it is forced to give up land, relocate farming, and the core of its regional economy. Indeed, there is a lot at stake for this actor that is beyond the scope of the model our policies are based on. Therefore, speaking truth to power is proven to not be enough.

Knowing this, we provide further recommendations to navigate the political arena. We try to address the arena from many actors' perspectives in order to provide a better understanding of the situation.

1. Rethinking the rational, speaking truth to power approach

The traditional rational approach used by Rijkswaterstaat, focused on finding the optimal solutions, is not suited for wicked problems characterized by deep uncertainty as mentioned before. This led to an oversimplification of the problem and reaching an outcome that lacked flexibility and was not desirable for all actors, such as Gelderland or Gorssel. To address this, a shift in approach is needed, which they tried to do a bit in the final debate, but should go further. A shift is needed towards an approach that embraces and systematically manages uncertainty and complexity. Rijkswaterstaat should prioritize looking for policies that are robust and that perform well in a wide range of possible futures (Maier et al., 2016). Policies should also be flexible and adaptive, through for example enabling adjustments over time. For example, an initial implementation of one Room for the River project in Gelderland could be monitored and after decisions could be adjusted based on performance. Instead of two, it could be limited to one project only. A method that could be used for this is Dynamic Adaptive Policy Pathways (DAP), which integrates an iterative process of planning and implementation that accounts for uncertainty (Walker et al., 2013). This involves designing an initial policy and adapting it dynamically when "trigger values" are reached through monitoring. This would allow for more effective management of uncertainty over time.

2. Foster co-creation and participatory modelling

Another limitation was the insufficient inclusion of diverse stakeholder perspectives. Gelderland's wishes could not be captured by the model, undermining their legitimacy and resulting in an outcome not favourable for them. And for the Transport Company, Rijkswaterstaat tried to ignore their priority of minimum water level, this was identified in previous sections as "excess of objectivity".

While Rijkswaterstaat aimed to increase transparency and understanding of perspectives by sharing models, this was not enough.

Therefore, Rijkswaterstaat should foster co-creation of knowledge. As a first step, they need to acknowledge diverse views and recognize the disagreements and avoid trying to resolve them through purely technical means. Then, co-creation can be done through participatory modeling approaches. This involves engaging stakeholders to jointly frame problems (Moallemi et al., 2020). This would have led to a better understanding of the position of Gelderland and the Transport Company and lead to a more inclusive policy.

After addressing the highest-level perspective, we now address the challengers position. We focus on the Transport Company and the Province of Gelderland.

3. Strengthening advocacy by the Transport Company

While the above is for the side of Rijkswaterstaat's improvements on not ignoring the water level, the TC could have also improved by better advocating for it. The TC should have reframed the minimum water level and leveraged it as a deep uncertainty (Moallemi et al., 2020), instead of presenting it as a simple priority. For example they could have used climate change impacts on the water level to show uncertainty. This moves the discussion from "known facts" to "shared unknowns", which would increase the likelihood of engagement by Rijkswaterstaat (Marchau et al., 2019).

4. Transparency and engagement for Gelderland and the TC

Both Gelderland and the TC adopted a strategy of secrecy and withholding information at the beginning of the process, which limited their influence in the outcome. To get more influence, both actors should pivot to an active, transparent and collaborative engagement strategy. Instead of challenging the policy from the outside at the end of the decision-making process, they should engage more fully in the process from the beginning. They should argue against a single "definitive" interpretation of scientific advice and demand for diverse implications.

Dealing with wicked problems and deep uncertainty, as seen in the Room for the River project, is inherently challenging. However, the recommendations presented above, could help foster a more robust, flexible and inclusive decision-making.

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